

Supply, Demand, and Market Equilibrium

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The core text is N. Gregory Mankiw, *Principles of Microeconomics*.

- **Chapter 4** — The Market Forces of Supply and Demand.

Read the chapter alongside these slides.

1. **Markets and competition** — these are the fundamentals.
2. **Demand** — we explain the law of demand, the equation, the table, the curve.
3. **From one buyer to the market** — where a market demand curve comes from.
4. **What shifts demand** — and why a shift is not a movement.
5. **Supply** — the same four steps, from the sellers' side.
6. **What shifts supply** — and the one shifter that runs backwards.
7. **Market equilibrium** — this is where the forces of supply and demand are in balance.
8. **Comparative statics** — explains what happens to equilibrium when things change.
9. **From table to equation to graph** — when you know the data first, not the equation.

Each part ends with a question for you to answer before we move on.

PART 1

Markets and Competition

The puzzle we are here to solve

How is the price of a cup of coffee actually determined?

Think about everything that happened before you paid it. Beans planted on a hillside a continent away. A growing season, and weather nobody controls. Pickers hired and paid. Sacks trucked to a port, freight across an ocean, roasting and blending and packing. Then a shop, a machine, a barista. And at the end of all of it, a price.

Thousands of people, none of whom know you, and no one setting the price.

What is a market?

Market. A market is a place or context where buyers and sellers trade. Buyers as a group determine demand; sellers as a group determine supply.

A market need not be a place. Crude oil trades through screens linking continents; consumers buy goods via internet platforms.

What makes it a market is the trading, not the geography.

Markets and competition

Market. A place or context where buyers and sellers meet to trade a particular good or service.

Buyers as a group determine market demand. Sellers as a group determine market supply.

Competition. Rivalry between agents over scarce resources.

Agents here can be individuals, firms, or governments. Buyers compete with each other for coffee. Sellers compete with each other for customers. Out of that rivalry a price emerges — and no individual buyer or seller plans it.

The market price is determined by the interaction of buyers and sellers.

Which of these is *not* a competitive market?

- A. Wheat farming in the US Midwest
- B. Corner coffee shops in Boston
- C. Tap water supply in a single city
- D. Second-hand textbooks on campus

60 seconds. Then explain your answer to the person next to you.

PART 2

Demand

Quantity demanded

Quantity demanded (Q^d). The amount of a good that buyers are willing and able to buy at a given price, in a given period.

Two words carry weight.

Willing rules out what people would like but will not pay for.

Able rules out what they would pay for but cannot afford.

And Q^d is a **flow**: cups per week, not cups.

The law of demand

Law of demand. Other things equal, when the price of a good rises, the quantity demanded of that good falls.

Hold on to the phrase **other things equal**. It is not decoration.

For now: freeze income, tastes, and every other price. Vary only the price of coffee.

Step 1 — write the equation

For our coffee market:

$$Q^d = 70 - 10P$$

Q^d in thousands of cups per week. P in dollars per cup.

The 70 — what buyers would take if coffee were free.

The -10 — each extra dollar cuts quantity demanded by 10 thousand cups. *The minus sign is the law of demand, written in algebra.*

Step 2 — plug the numbers in

Price P (\$/cup)	Substitute into $Q^d = 70 - 10P$	Q^d (000s/week)
1	$70 - 10(1)$	= 60
2	$70 - 10(2)$	= 50
3	$70 - 10(3)$	= 40
4	$70 - 10(4)$	= 30
5	$70 - 10(5)$	= 20
6	$70 - 10(6)$	= 10
7	$70 - 10(7)$	= 0

The table is not a separate object. **It is the equation, evaluated seven times.**

One row worth pausing on

At $P = 7$, quantity demanded is **zero**. The price is so high that nobody buys at all.

Call this the choke price.

It is where the demand curve meets the price axis, and you can read it straight off the equation: set $Q^d = 0$ and solve for P .

Step 3 — rearrange to plot

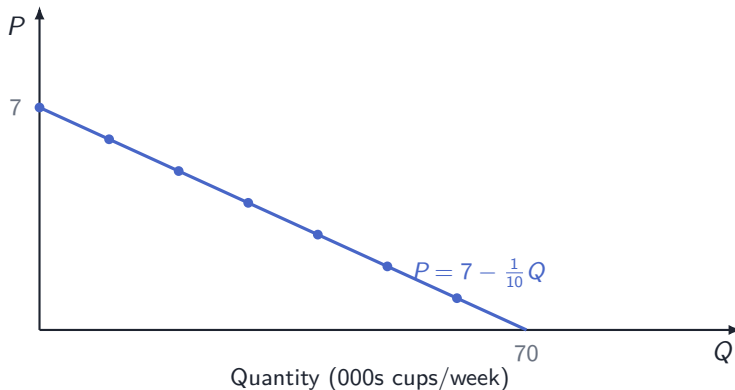
Economists put price on the *vertical* axis. That is backwards from the mathematics you were taught — a habit inherited from Alfred Marshall.

So rearrange to make P the subject:

$$Q^d = 70 - 10P \implies 10P = 70 - Q^d \implies P = 7 - \frac{1}{10}Q^d$$

Same relationship. Same information. Different subject.

Step 4 — draw the graph



The dots are the seven rows. The line is the function. **They coincide because they are the same object.**

Active Learning 2

Demand for tea is $Q^d = 90 - 15P$

1. What is quantity demanded at $P = 4$?
2. What is the choke price?
3. If the price rises from \$2 to \$3, by how much does Q^d fall?

On paper. Three minutes. Show your substitution.

PART 3

From One Buyer to the Market

Where did that curve come from?

We wrote down a *market* demand curve and never asked where it came from.

A market demand curve is built from individual demand curves.

Each buyer has their own. To get the market curve, add up the quantities **at each price**.

Two buyers

Suppose the market has just two buyers, Ama and Ben:

$$Q_{\text{Ama}}^d = 42 - 6P, \quad Q_{\text{Ben}}^d = 28 - 4P$$

Ama buys more at every price below \$7 and reacts more strongly to price changes (slope -6 against -4).

Both stop buying at \$7 — they happen to share a choke price.

Add the quantities at each price

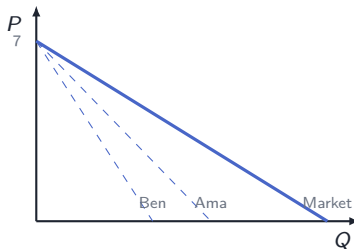
P	$A_{ma} = 42 - 6P$	$B_{en} = 28 - 4P$	Sum	Market Q^d
1	36	24	$36 + 24 = 60$	60
2	30	20	$30 + 20 = 50$	50
3	24	16	$24 + 16 = 40$	40
4	18	12	$18 + 12 = 30$	30
5	12	8	$12 + 8 = 20$	20
6	6	4	$6 + 4 = 10$	10
7	0	0	$0 + 0 = 0$	0

The last column is the market demand curve we started with.

In algebra, and on the graph

$$\underbrace{(42 - 6P)}_{\text{Ama}} + \underbrace{(28 - 4P)}_{\text{Ben}} = 70 - 10P$$

Intercepts add: $42 + 28 = 70$. Slopes add: $6 + 4 = 10$.



Active Learning 3

A market has two buyers:

$$Q_1^d = 20 - 2P, \quad Q_2^d = 30 - 3P$$

1. Write the market demand equation.
2. What is market quantity demanded at $P = 4$?
3. Check your answer by computing each buyer separately and adding.

On paper. Three minutes. Question 3 is the check — do not skip it.

PART 4

What Shifts Demand

We froze the world. Now unfreeze it.

The curve was drawn holding everything except the price of coffee constant.

Release one of those things and **the curve itself moves**.

Which things, in which direction, and why?

The intercept is not a fact about coffee

Write demand out with the other influences visible:

$$Q^d = 50 - 10P + 6P_T - 4P_S + 2Y$$

P_T = price of tea P_S = price of sugar Y = income index

At today's values — tea \$3, sugar \$2, income 5:

$$50 + 6(3) - 4(2) + 2(5) = 50 + 18 - 8 + 10 = 70$$

The rule in one line

Changing P moves you *along* the curve — P sits in the equation multiplied by its own slope.

Changing P_T , P_S **or** Y changes the *intercept*, which relocates the whole curve.

That is the entire distinction, and you can see it in the algebra.

Substitutes. Two goods that serve the same purpose, so that a rise in the price of one increases demand for the other. The coefficient on the other good's price is *positive*.

Worked example. A blight hits the tea crop; tea goes \$3 → \$5.

$$50 + 6(5) - 4(2) + 2(5) = 82 \quad \Rightarrow \quad Q^d = 82 - 10P$$

Demand rose by 12 thousand cups *at every price* — and the price of coffee never changed.

Complements. Two goods consumed together, so that a rise in the price of one decreases demand for the other. The coefficient is *negative*.

Worked example. Sugar goes \$2 $\rightarrow$ \$4.

$$50 + 6(3) - 4(4) + 2(5) = 62 \quad \Rightarrow \quad Q^d = 62 - 10P$$

Demand fell by 8 thousand cups at every price.

The complement chain, stated slowly

Sugar and coffee are consumed **together**. What a buyer really chooses is *the pair*.

When sugar becomes dearer, **the pair** becomes dearer — even though coffee itself has not changed price at all.

Buyers buy fewer coffee-and-sugar combinations, which means less coffee.

Demand for coffee shifts left.

Income: normal and inferior goods

Normal good. Demand increases when income rises. The coefficient on Y is positive.

Inferior good. Demand decreases when income rises. The coefficient on Y is negative.

Ours is $+2$, so coffee is normal here. Income $5 \rightarrow 10$ gives an intercept of 80.

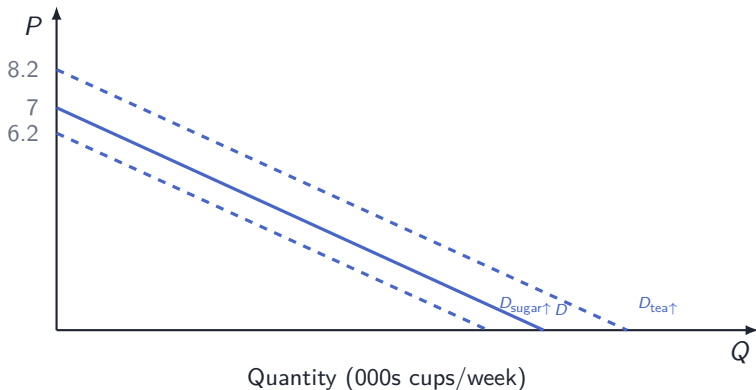
Inferior does not mean bad. It means people buy less of it as they get richer: instant coffee, coach travel, own-brand cereal.

The full list of demand shifters

Change	Effect on demand	Curve
Income rises (normal good)	increases	shifts right
Income rises (inferior good)	decreases	shifts left
Price of a substitute rises	increases	shifts right
Price of a complement rises	decreases	shifts left
Tastes turn toward the good	increases	shifts right
More buyers enter the market	increases	shifts right
Buyers expect a higher future price	increases	shifts right

The last row is the only forward-looking one: expectations shift demand *now*, on a price change that has not happened yet.

The three curves together



Intercepts 62, 70, 82. **The slope -10 never changes** — which is exactly what makes these shifts rather than movements.

Active Learning 4

For each event, say whether the **demand curve for coffee** shifts left, shifts right, or does not move — and *why*.

1. The price of coffee falls by \$1.
2. A recession cuts incomes, and coffee is a normal good.
3. The price of tea falls sharply.
4. A study reports that coffee improves memory.

Two minutes. The *why* is the marks.

PART 5

Supply

Quantity supplied

Quantity supplied (Q^s). The amount of a good that sellers are willing and able to sell at a given price, in a given period.

Law of supply. Other things equal, when the price of a good rises, the quantity supplied of that good rises.

Note it runs the *opposite* way to demand.

Why supply slopes upward

The reasoning runs through **cost**.

Producing more coffee means longer hours, extra staff, beans from less convenient places.

All of that costs more per cup.

A higher price is what makes those extra costs worth bearing.

Step 1 — write the equation

$$Q^s = 10P - 10$$

The +10 — each extra dollar brings out another 10 thousand cups. Positive, because that is the law of supply.

But the intercept is -10 . What is a negative quantity?

Reading the negative intercept

It does **not** mean sellers supply minus ten thousand cups at a price of zero. That would be meaningless.

It means the curve reaches the axis at a *positive price*. Set $Q^s = 0$:

$$10P - 10 = 0 \implies P = 1$$

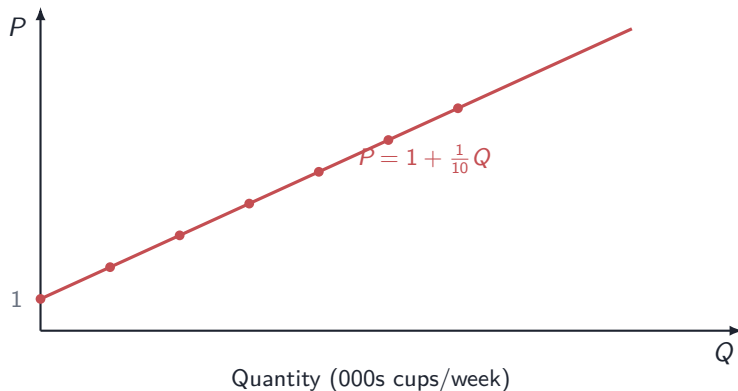
Below \$1 a cup, nothing is supplied at all. Call \$1 the **shut-down price**.

Step 2 — plug the numbers in

Price P (\$/cup)	Substitute into $Q^s = 10P - 10$		Q^s (000s/week)
1	$10(1) - 10$	=	0
2	$10(2) - 10$	=	10
3	$10(3) - 10$	=	20
4	$10(4) - 10$	=	30
5	$10(5) - 10$	=	40
6	$10(6) - 10$	=	50
7	$10(7) - 10$	=	60

Step 3 — rearrange to plot: $P = 1 + \frac{1}{10} Q^s$

Step 4 — draw the graph



The curve meets the price axis at \$1 — the shut-down price. Below it, the curve does not exist.

Supply of tea is $Q^s = 12P - 24$

1. What is quantity supplied at $P = 5$?
2. What is the shut-down price?
3. At $P = 1$, how much is supplied? Explain your answer in words.

On paper. Three minutes. Question 3 is the one that matters.

PART 6

What Shifts Supply

The same logic, mirrored

The intercept -10 is not a primitive either. It is assembled from the conditions sellers face:

$$Q^S = 10P + 16 - 5P_B + 4T$$

P_B = price of beans (an input) T = index of roasting technology

At beans \$6, technology 1:

$$16 - 5(6) + 4(1) = 16 - 30 + 4 = -10$$

Read the two signs

Beans, -5: negative, because beans are a *cost*. When an input gets dearer, the margin on each cup falls, the least efficient sellers stop producing, and supply shifts **left**.

Every input behaves this way — wages, rent, energy, shipping.

Technology, +4: positive, because better technology cuts the cost of each cup. Same beans, same hours, more output. Supply shifts **right**.

Two worked examples

A poor harvest. Beans \$6 $\rightarrow$ \$8:

$$16 - 5(8) + 4(1) = -20 \quad \Rightarrow \quad Q^s = 10P - 20$$

Supply fell by 10 at every price, *and* the shut-down price rose \$1 $\rightarrow$ \$2.

Better technology. Index 1 $\rightarrow$ 3:

$$16 - 5(6) + 4(3) = -2 \quad \Rightarrow \quad Q^s = 10P - 2$$

Supply rose by 8 at every price; the shut-down price fell to \$0.20.

The full list of supply shifters

Change	Effect on supply	Curve
Input prices rise (beans, wages, energy)	decreases	shifts left
Technology improves	increases	shifts right
More sellers enter the market	increases	shifts right
Sellers leave the market	decreases	shifts left
A tax per unit is imposed	decreases	shifts left
A subsidy per unit is paid	increases	shifts right
Sellers expect a higher future price	decreases	shifts left
Bad harvest, drought or disruption	decreases	shifts left

Expectations run the other way

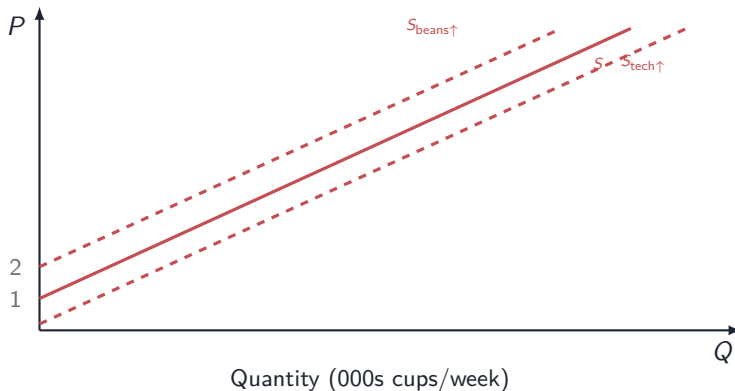
For *buyers*, expecting a higher price next month means buying **more today** — demand shifts right.

For *sellers*, expecting a higher price next month means holding stock back — supply shifts **left**.

Buyers pull purchases forward. Sellers push sales back.

Both require a good that can be stored, or a purchase that can be moved in time. Nobody holds back a brewed coffee.

Two supply shifts on one diagram



Parallel again — the price coefficient $+10$ never changes. Only how much sellers will supply *at* each price.

Active Learning 6

For each event, does the **supply curve for coffee** shift left, shift right, or not move — and *why*?

1. The minimum wage rises.
2. The price of coffee rises by \$1.
3. New roasting machinery halves energy use per cup.
4. Twenty new coffee roasters enter the market.

Two minutes. The *why* is the marks.

PART 7

Equilibrium

Putting the two halves together

Equilibrium. The price and quantity at which quantity demanded equals quantity supplied. The market clears: every buyer willing to pay that price finds a seller, and every seller willing to accept it finds a buyer.

Two curves. One price where they meet.

Solving for equilibrium

Set $Q^d = Q^s$ and solve:

$$70 - 10P = 10P - 10$$

$$70 + 10 = 10P + 10P$$

$$80 = 20P$$

$$P^* = 4$$

Always check

Substitute back using **both** equations:

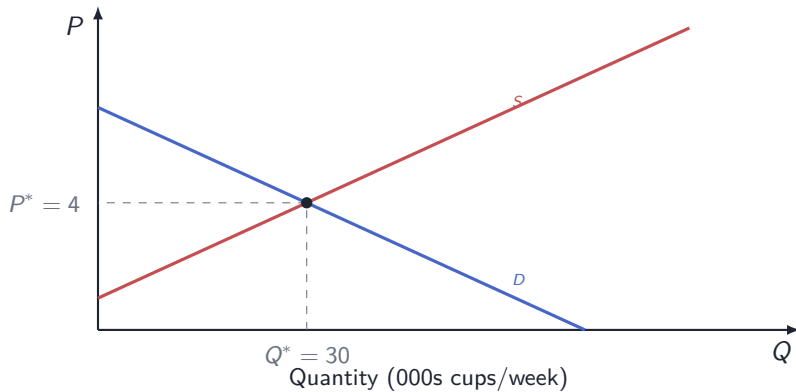
$$Q^d = 70 - 10(4) = 30, \quad Q^s = 10(4) - 10 = 30$$

Both give 30. That is your check.

Equilibrium: \$4 per cup, 30 thousand cups per week.

If they had disagreed, you would know to go looking.

Equilibrium on the diagram



Why the market gets there

Shortage. $Q^d > Q^s$. Buyers want more than sellers will provide. Occurs when the price is *below* equilibrium.

Surplus. $Q^s > Q^d$. Sellers offer more than buyers will take. Occurs when the price is *above* equilibrium.

A shortage at \$3

$$Q^d = 70 - 10(3) = 40, \quad Q^s = 10(3) - 10 = 20, \quad \text{shortage} = 20$$

Queues form. Sellers notice they can raise prices without losing customers — and they do.

As price rises, Q^d falls *and* Q^s rises.

The gap closes from both sides, and stops exactly at \$4.

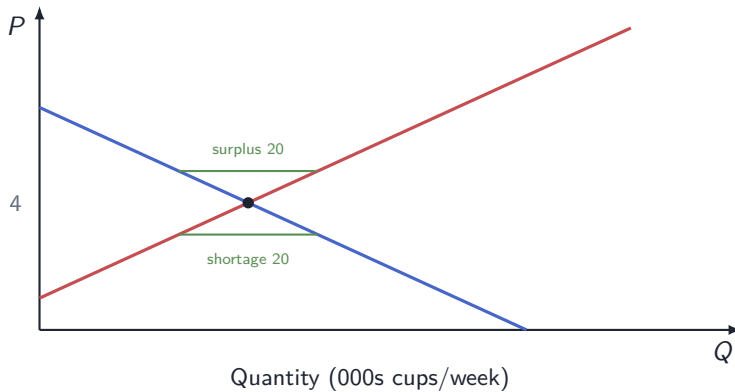
A surplus at \$5

$$Q^d = 70 - 10(5) = 20, \quad Q^s = 10(5) - 10 = 40, \quad \text{surplus} = 20$$

Unsold coffee. Sellers cut prices to shift it, Q^s falls, Q^d rises.

Again the market lands at \$4. It is self-correcting from either side.

Both on one diagram



Active Learning 7

A market for umbrellas:

$$Q^d = 50 - 5P, \quad Q^s = 10P - 25$$

1. Solve for P^* and Q^* .
2. Check your Q^* using *both* equations.
3. At $P = 3$, is there a shortage or a surplus? How large?

On paper. Four minutes.

PART 8

Comparative Statics

What the model is actually for

Comparative statics. Comparing the equilibrium before an event with the equilibrium after it. It answers “what changed”, not “how long did it take”.

Everything so far has been machinery. **This is the part you use.**

The three-step method

Every question of this kind is answered the same way.

1. Decide **which curve shifts** — demand, supply, or both.
2. Decide **which direction**, and change the intercept accordingly.
3. **Solve** the new pair of equations. Read off P^* and Q^* .

Learn the method, not the answers.

Case 1: demand increases

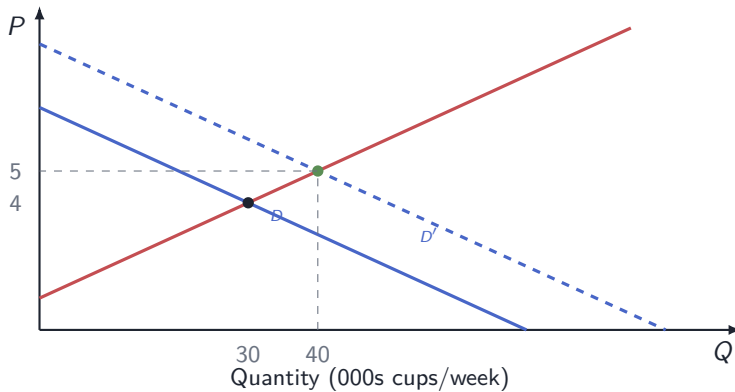
A study reports that coffee is good for you. Demand rises by 20 at every price.

New demand	$Q^d = 90 - 10P$
Supply	$Q^s = 10P - 10$ (unchanged)

Set equal	$90 - 10P = 10P - 10$
	$100 = 20P$
	$P^* = 5$
Substitute	$Q^* = 10(5) - 10 = 40$

Price \$4 $\rightarrow$ \$5. Quantity 30 $\rightarrow$ 40. Both rise.

Case 1 on the diagram



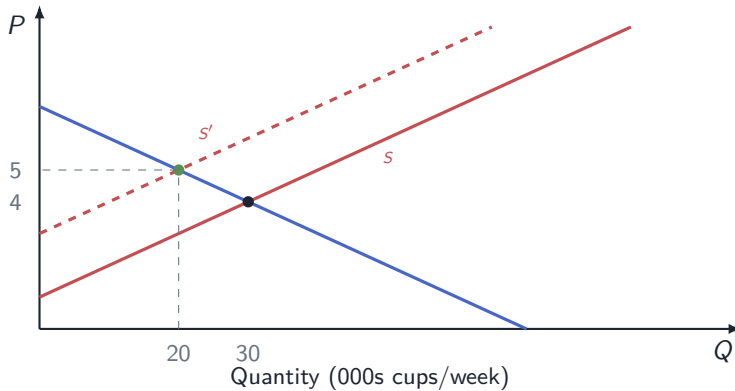
Case 2: supply decreases

A frost destroys part of the Brazilian crop. Supply falls by 20 at every price.

Demand	$Q^d = 70 - 10P$ (unchanged)
New supply	$Q^s = 10P - 30$
Set equal	$70 - 10P = 10P - 30$ $100 = 20P$ $P^* = 5$
Substitute	$Q^* = 10(5) - 30 = 20$

Price \$4 → \$5. Quantity 30 → 20. They move apart.

Case 2 on the diagram



Compare the two cases

Both raised the price to exactly \$5.

A student who *only looked at the price* could not tell them apart.

The quantity distinguishes them: it rose in Case 1, fell in Case 2.

Price alone never identifies which curve moved.

Case 3: both curves shift

Demand rises by 20 *and* supply falls by 20.

$$90 - 10P = 10P - 30 \Rightarrow 120 = 20P \Rightarrow P^* = 6, \quad Q^* = 30$$

Price rises to \$6. Quantity is 30 — **exactly where it started**.

Both shifts push price the same way, so price is unambiguous. They push quantity in *opposite* directions, so quantity depends on which shift is bigger.

Active Learning 8

Start from $Q^d = 70 - 10P$ and $Q^s = 10P - 10$ ($P^* = 4$, $Q^* = 30$).

A shipping crisis cuts supply by **40** at every price.

1. Write the new supply equation.
2. Find the new P^* and Q^* .
3. Before calculating: which way did you *expect* each to move?

Four minutes. Do question 3 *first*.

PART 9

From Table to Equation to Graph

We have been doing this backwards

All lecture, I gave you an equation and you produced the table and the graph.

Nobody hands you an equation.

Data arrives as a **table**: prices, and the quantities observed at those prices.

The equation is something *you* have to find.

The data

A market for bicycle helmets. Observed at five prices:

Price P (\$)	1	2	3	4	5
Quantity demanded Q^d (000s/week)	90	80	70	60	50
Quantity supplied Q^s (000s/week)	0	20	40	60	80

That is all you get. **Four steps get you from here to a full model.**

Step 1 — find the slope: rise over run

Pick *any* two rows of the demand data. Take $P = 1$ and $P = 2$:

$$\text{slope} = \frac{\text{rise}}{\text{run}} = \frac{\text{change in } Q^d}{\text{change in } P} = \frac{80 - 90}{2 - 1} = \frac{-10}{1} = -10$$

Each extra dollar reduces quantity demanded by 10 thousand helmets.

The minus sign is the law of demand, arriving from the data rather than from theory.

Any two rows give the same answer

Rows used	Rise over run	Slope
$P = 1$ and $P = 2$	$(80 - 90)/(2 - 1)$	-10
$P = 2$ and $P = 4$	$(60 - 80)/(4 - 2)$	-10
$P = 1$ and $P = 5$	$(50 - 90)/(5 - 1)$	-10

The same slope every time. That is what makes the relationship linear.

If different pairs of rows gave different slopes, the curve would be bent — and a straight-line equation would be the wrong model.

Step 2 — find the intercept

We know the shape: $Q^d = a - 10P$. One unknown left: a .

Substitute any row you like. Take $P = 1$, $Q^d = 90$:

$$90 = a - 10(1)$$

$$90 = a - 10$$

$$a = 100$$

So $Q^d = 100 - 10P$.

Step 3 — check against a row you did not use

We built the equation from the $P = 1$ row. So test it on a different one.

At $P = 4$: $Q^d = 100 - 10(4) = 60$. Table says **60**. ✓

At $P = 5$: $Q^d = 100 - 10(5) = 50$. Table says **50**. ✓

Never accept an equation you fitted without testing it on data you did not use to fit it.

Now do the same for supply

Slope. From $P = 1$ and $P = 2$: $\frac{20 - 0}{2 - 1} = +20$ — positive, as the law of supply requires.

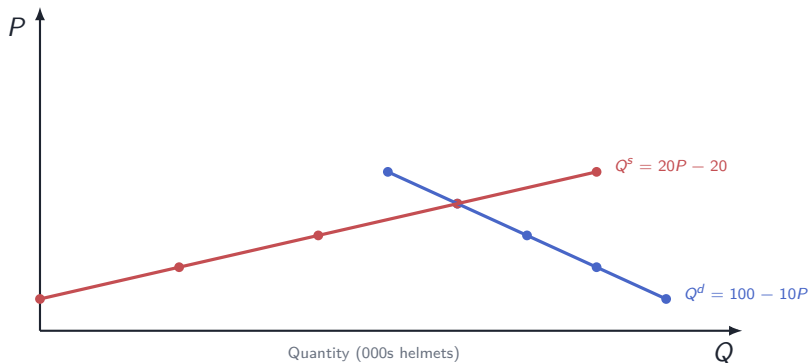
Intercept. $Q^s = c + 20P$. At $P = 1$, $Q^s = 0$:

$$0 = c + 20(1) \implies c = -20$$

So $Q^s = 20P - 20$.

Check at $P = 5$: $20(5) - 20 = 80$. Table says **80**. ✓

Step 4 — plot both from the table



Dots are the ten table entries. Lines are the two equations. **They coincide.**

Step 5 — equilibrium, two ways

From the table. Look down the two rows for a price where the numbers match. At $P = 4$: $Q^d = 60$ and $Q^s = 60$.

From the equations.

$$100 - 10P = 20P - 20$$

$$120 = 30P$$

$$P^* = 4, \quad Q^* = 100 - 10(4) = 60$$

Both routes give \$4 and 60 thousand. **Two methods, one answer.**

Why the equations beat the table

The table gave us the equilibrium here only because \$4 *happened* to be one of the five rows.

It usually will not be.

The equations work at every price, observed or not. They give you the choke price of \$10 — which appears nowhere in the data, and follows only from extending the fitted line past the prices we were given.

That is what you gain by turning a table into a model.

Active Learning 9

A market for umbrellas. This is all the data you have:

Price P (\$)	2	4	6
Q^d (000s/week)	60	40	20
Q^s (000s/week)	0	20	40

1. Find both slopes by rise over run.
2. Find both intercepts, and write both equations.
3. Check each equation against a row you did not use.
4. Find P^* and Q^* .

On paper. Six minutes. Question 4 has a trap in it.

Nobody decides the price. Everybody does.

Two schedules meet, and out of them come a price and a quantity carrying information no one person holds. What follows is about when that mechanism works, and when it does not.

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